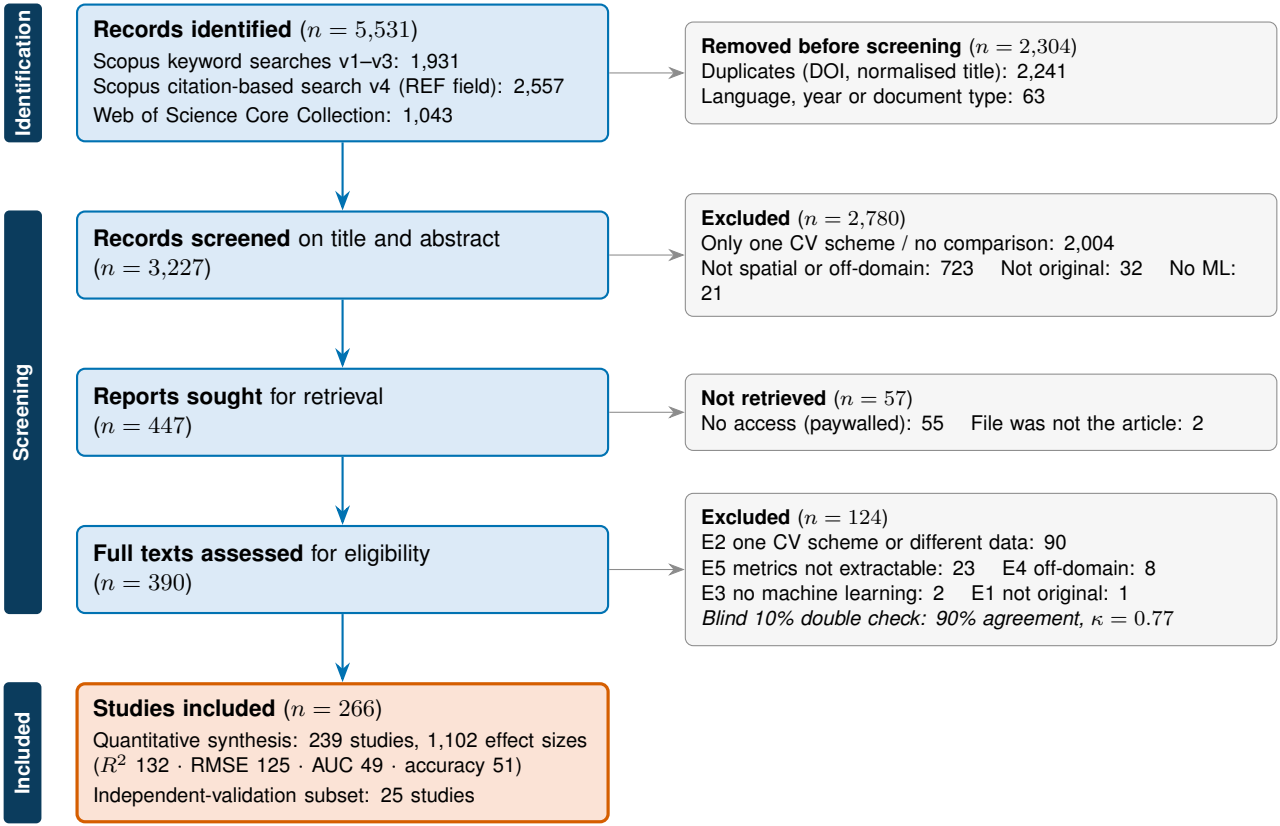

Identification
Records identified ($n = 5{,}531$)
Scopus keyword searches v1–v3: 1,931
Scopus citation-based search v4 (REF field): 2,557
Web of Science Core Collection: 1,043
Removed before screening ($n = 2{,}304$)
Duplicates (DOI, normalised title): 2,241
Language, year or document type: 63
Screening
Records screened on title and abstract ($n = 3{,}227$)
Excluded ($n = 2{,}780$)
Only one CV scheme / no comparison: 2,004
Not spatial or off-domain: 723 Not original: 32 No ML: 21
Reports sought for retrieval ($n = 447$)
Not retrieved ($n = 57$)
No access (paywalled): 55 File was not the article: 2
Full texts assessed for eligibility ($n = 390$)
Excluded ($n = 124$)
E2 one CV scheme or different data: 90
E5 metrics not extractable: 23 E4 off-domain: 8
E3 no machine learning: 2 E1 not original: 1
Blind 10% double check: 90% agreement, $\kappa = 0.77$
Included
Studies included ($n = 266$)
Quantitative synthesis: 239 studies, 1,102 effect sizes
(R^2 132 · RMSE 125 · AUC 49 · accuracy 51)
Independent-validation subset: 25 studies